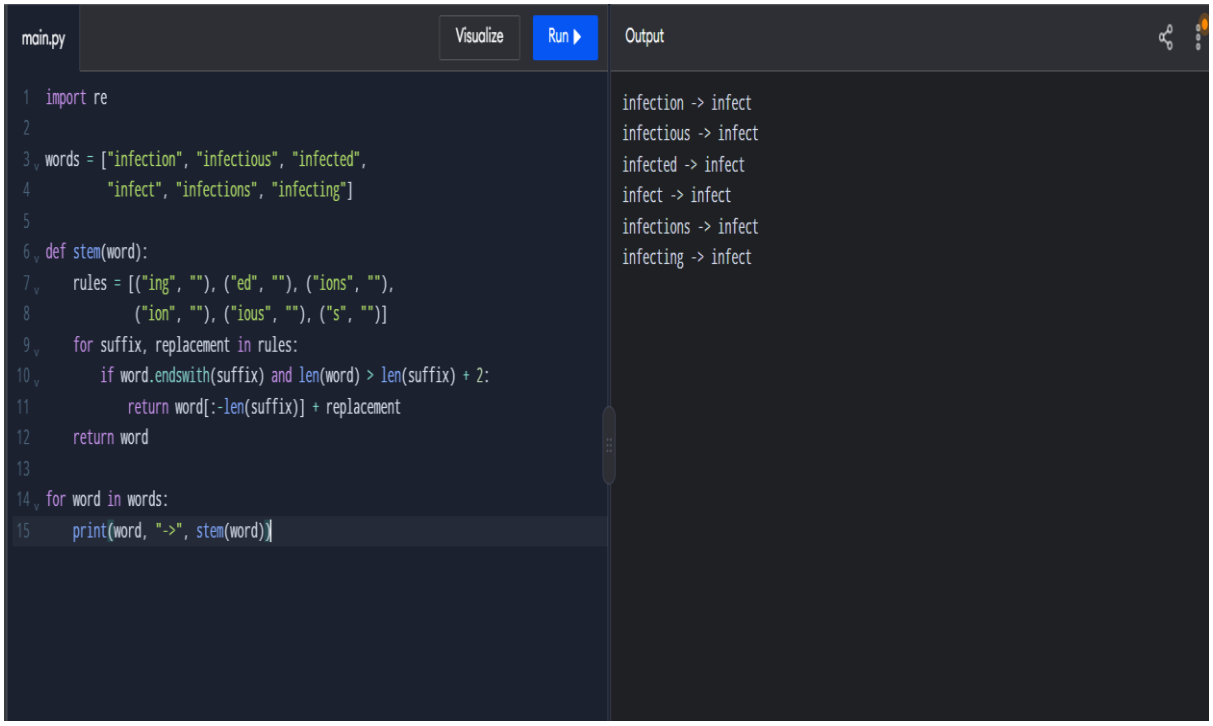
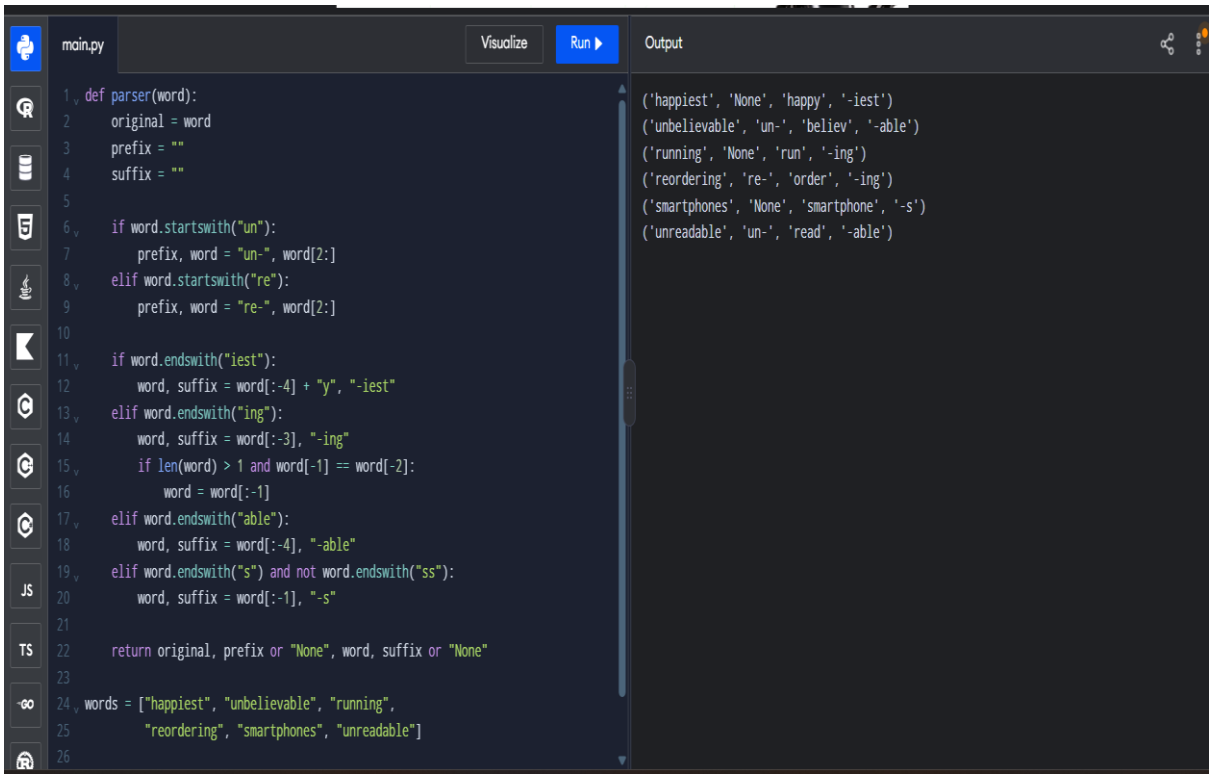
NAME: VIJAYALAKSHMI S

REG NO:192472326

Q1)



Q2)



Q3)

main.py

```python
import re
import time
from collections import Counter

documents = [
    "organization organizer organizing organized",
    "technology technology business business",
    "news reports organizing organizations",
    "business technology organizer organization"
]

def simple_stem(word):
    for suffix in ["ing", "ed", "er", "tion", "s"]:
        if word.endswith(suffix) and len(word) > len(suffix) + 2:
            return word[:-len(suffix)]
    return word

def simple_lemma(word):
    rules = {
        "organized": "organize",
        "organizing": "organize",
        "organizer": "organizer",
        "organizations": "organization"
    }
    return rules.get(word, word)
```

Output

```
none Vocabulary: 9 Time: 0.000117 seconds
stem Vocabulary: 7 Time: 3.3e-05 seconds
lemma Vocabulary: 7 Time: 2e-05 seconds
```

Q4)

main.py

```python
words = ["watches", "watching", "washable", "washer", "washed"]

rules = {
    "watches": ("watch", "Inflectional plural"),
    "watching": ("watch", "Inflectional"),
    "washable": ("washable", "Derivational"),
    "washer": ("washer", "Derivational"),
    "washed": ("wash", "Inflectional past tense")
}

for word in words:
    root, category = rules[word]
    print(word, "->", root, "|", category)
```

Output

```
watches -> watch | Inflectional plural
watching -> watch | Inflectional
washable -> washable | Derivational
washer -> washer | Derivational
washed -> wash | Inflectional past tense
```

Q5)

main.py

```python

import re

def stem(word):
    for suffix in ["ing", "ed", "ly", "es", "s"]:
        if word.endswith(suffix) and len(word) > len(suffix) + 2:
            return word[:-len(suffix)]
    return word

documents = [
    "The government announced new policies.",
    "Technology companies organized a conference.",
    "The organizers discussed business development.",
    "Researchers are studying new technologies.",
    "The organization published its annual report."
]

for text in documents:
    words = re.findall(r"[a-zA-Z]+", text.lower())
    processed = [stem(word) for word in words]

    print("\nOriginal:", text)
    print("Stemmed :", " ".join(processed))

```

Output

```
Original: The government announced new policies.
Stemmed : the government announc new polici

Original: Technology companies organized a conference.
Stemmed : technology compani organiz a conference

Original: The organizers discussed business development.
Stemmed : the organizer discuss busines development

Original: Researchers are studying new technologies.
Stemmed : researcher are study new technologi

Original: The organization published its annual report.
Stemmed : the organization publish its annual report
```

Q6)

main.py

```python
import re
import time
from collections import Counter

documents = [
    "organization organizer organizing organized",
    "technology technology business business",
    "news reports organizing organizations",
    "business technology organizer organization"
]

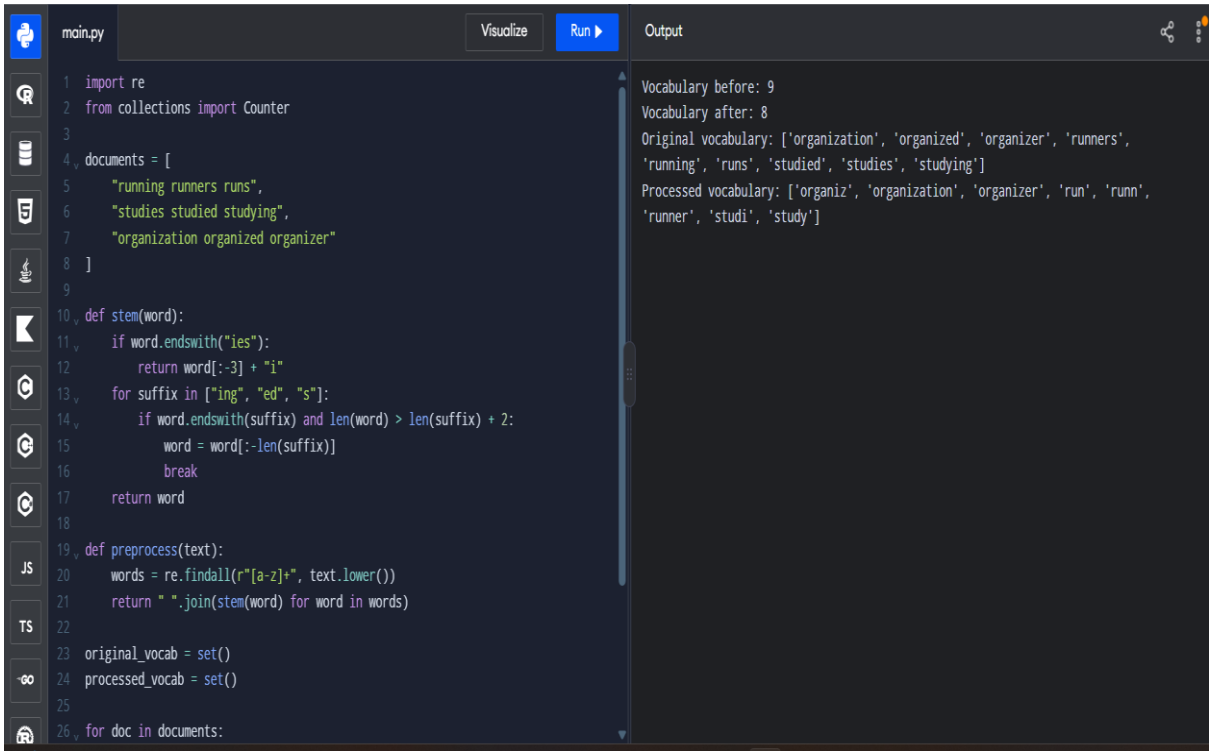
def simple_stem(word):
    for suffix in ["ing", "ed", "er", "tion", "s"]:
        if word.endswith(suffix) and len(word) > len(suffix) + 2:
            return word[:-len(suffix)]
    return word

def simple_lemma(word):
    rules = {
        "organized": "organize",
        "organizing": "organize",
        "organizer": "organizer",
        "organizations": "organization"
    }
    return rules.get(word, word)
```

Output

```
none Vocabulary: 9 Time: 0.000117 seconds
stem Vocabulary: 7 Time: 3.3e-05 seconds
lemma Vocabulary: 7 Time: 2e-05 seconds
```

Q7)



main.py

Visualize

Run

```
import re
from collections import Counter

documents = [
    "running runners runs",
    "studies studied studying",
    "organization organized organizer"
]

def stem(word):
    if word.endswith("ies"):
        return word[:-3] + "i"
    for suffix in ["ing", "ed", "s"]:
        if word.endswith(suffix) and len(word) > len(suffix) + 2:
            word = word[:-len(suffix)]
            break
    return word

def preprocess(text):
    words = re.findall(r"[a-z]+", text.lower())
    return " ".join(stem(word) for word in words)

original_vocab = set()
processed_vocab = set()

for doc in documents:
```

Output

```
Vocabulary before: 9
Vocabulary after: 8
Original vocabulary: ['organization', 'organized', 'organizer', 'runners',
'running', 'runs', 'studied', 'studies', 'studying']
Processed vocabulary: ['organiz', 'organization', 'organizer', 'run', 'runn',
'runner', 'studi', 'study']
```